

ReFLIP-VAD: Towards Weakly Supervised Video Anomaly Detection via Vision-Language Model

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Abstract—The vision-language model has recently achieved notable success in image-related tasks, showcasing its ability to learn deep and meaningful visual representations. Applying this robust model to video analysis for detecting anomalies poses a significant challenge. This paper introduces Reparameterized Fine-grained Language Image Pretraining-Video Anomaly Detection (ReFLIP-VAD), a novel approach designed to leverage vision-language capabilities for video anomaly detection. ReFLIP-VAD employs a prompt encoder to generate reparameterized learnable prompt templates, enhancing interpretability and understanding of anomaly-specific semantics. The framework adopts a dual-block architecture: a classification block that uses visual features for binary classification and a video-text alignment block that integrates textual and visual features for precise language-vision alignment. This proposed approach is further strengthened by the Glimpse-Emphasize network that effectively captures both global and local temporal dependencies across time and the MIL-Align mechanism that selects the most representative video frames for each label, representing the entire video. ReFLIP-VAD has demonstrated superior performance on two large-scale benchmark datasets achieving an Average Precision (AP) of 86.29% on XD-Violence and an Area Under the Curve (AUC) of 89.14% on UCF-Crime, significantly surpassing existing state-of-the-art methods. Our source code is available at <https://github.com/prasaddev97/ReFLIP-VAD>.

Index Terms—Video anomaly detection, Weakly supervised, Vision-language pre-training, Prompt template

I. INTRODUCTION

VIDEO anomaly detection (VAD) task has recently gained significant attention in real-world scenarios, including but not limited to public surveillance, transportation management, traffic monitoring, and military settings. This task is approached through three primary paradigms, namely, Fully-Supervised VAD (FSVAD) [1], Unsupervised VAD (USVAD) [2], and Weakly-Supervised VAD (WSVAD) [3]. Traditional FSVAD is deemed impractical due to the scattered and diverse anomalies that require extensive manual frame-level annotations. Conversely, unsupervised VAD, which relies solely on learning from normal videos to identify open-set anomalies, often generates false alarms. However, it is inherently challenging to determine what is normal and abnormal without prior knowledge when provided only with normal videos. In

contrast to both FSVAD and USVAD, our primary focus is on the practical paradigm of WSVAD, which utilizes only video-level annotations. This paradigm results in reduced costs for manual and detailed annotations.

Current research in the VAD domain typically follows a systematic approach. It starts with extracting only visual features using visual backbone models, e.g., C3D [4], [5], I3D [6], [7], ViT [8], which are pre-trained on activity recognition tasks. These extracted features are then fed into binary classifiers based on multiple instance learning (MIL) framework [9]. The last step involves detecting abnormal events using the predicted anomaly confidence. Although these straightforward approaches have shown potential results, they do not fully harness cross-modal connections between vision and language.

In recent years, significant advancements have been observed in the advancement of vision-language models such as Contrastive Language-Image Pre-training (CLIP) [10], A Large-scale Image and Noisy-text embedding (ALIGN) [11], Context Optimization (CoOp) [12] and Fine-grained Language-Image Pre-training (FLIP) [13]. These models aim to acquire generalized visual representations enriched with semantic concepts. The key idea behind CLIP is to align images and text in a shared embedding space using contrastive learning, allowing it to associate images with their textual descriptions and perform various tasks involving vision and language. Given the remarkable success of VLMs in recent times, there is a growing interest in exploring the development of task-specific models built upon CLIP.

Despite the evident potential of VLMs in diverse visual tasks, their primary emphasis lies within the realm of static images. Hence, it is imperative to thoroughly investigate how to effectively transform a model trained on image-text pairs into a tool capable of addressing the more intricate challenge of video anomaly detection with limited supervision. To harness generalized knowledge effectively and enable FLIP to realize its full potential in the WSVAD task, it is crucial to address specific challenges aligned with the characteristics of WSVAD. Firstly, exploring methods to capture contextual dependencies across time is vital. Secondly, determining how to utilize acquired knowledge and enhance visual-language connections is essential. Thirdly, maintaining vision-language model performance in video anomaly detection scenarios is of critical importance.

To effectively address the above-mentioned issues, a novel framework **ReFLIP-VAD** is developed: **Re**parameterized **F**ine-grained **L**anguage **I**mage **P**re-training for **V**ideo **A**nomaly **D**etection. A Glimpse-Emphasize network is introduced to address the first challenge based on a global-to-local strat-

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egy. It comprises two key components: the Glimpse module and the Emphasize module. The Glimpse module primarily captures the global context information and the Emphasize module extracts the local features. Addressing the second challenge divergently from existing approaches that only rely on visual features, ReFLIP-VAD incorporates textual features with visual features. This multi-modality strategy aims to maximize the preservation of learned knowledge. ReFLIP-VAD involves a classification block and a video-text alignment block. The former leverages visual features for binary classification while the latter leverages both textual and visual features for language-vision alignment. Furthermore, these blocks effortlessly accomplish both coarse-grained and fine-grained weakly supervised video anomaly detection [14]. In addition to these blocks, two prompt mechanisms namely the multi-modal prompts and reparameterized learnable prompts are introduced to enhance text label representations. Contrary to existing methods [10], [12], [15], [16], which depend on either hand-crafted or learnable prompts, the proposed approach utilizes a prompt encoder, specifically DistilBERT [17], to generate reparameterized learnable prompt template. These templates are contextually rich and provide a level of interpretability, facilitating a more detailed understanding of the specific semantics associated with anomalies. To address the third challenge, a multiple instance learning (MIL)-align mechanism is utilized for the visual and language alignment block.

The main contributions of this paper are outlined as follows:

- 1) A novel framework i.e. ReFLIP-VAD is developed that employs a prompt encoder to generate reparameterized learnable prompt templates instead of hand-crafted templates. These templates are contextually rich, enhancing interpretability and providing a deeper understanding of the specific semantics associated with anomalies.
- 2) The proposed approach comprises a classification block and a video-text alignment block. The former leverages visual features for binary classification while the latter leverages both textual and visual features for language-vision alignment. Consequently, this dual block based proposed approach is able to detect video anomalies at both coarse and fine-grained levels.
- 3) A Glimpse-Emphasize network is developed that effectively captures both the global and local temporal dependencies across time. The MIL-Align mechanism is also developed to optimize visual-language alignment under weak supervision.
- 4) The effectiveness of ReFLIP-VAD is demonstrated on two large-scale benchmarks. ReFLIP-VAD achieves state-of-the-art performance, including 86.29% AP on XD-Violence and 89.14% AUC on UCF-Crime, surpassing existing state-of-the-art methods by a large margin.

The rest of the paper is structured as follows: Section II discusses the state-of-the-art approaches for the detection of anomalies in videos and how to adapt vision-language models in video anomaly detection. Section III describes the proposed methodology that achieves both coarse-grained and fine-grained video anomaly detection effectively. Section IV

presents the experimental results and discussion on the UCF Crime and XD-Violence dataset. Section V concludes with a discussion of future research directions.

II. RELATED WORK

A. Video Anomaly Detection

1) *Unsupervised VAD (USVAD)*: Video anomaly detection has undergone a revolutionary transformation with the advent of deep learning techniques. Methods utilizing convolutional neural networks [18]–[20], recurrent neural networks [21], [22], and transformers [23], [24] have progressively emerged as the primary focus of research in this area. Peng et al. [18] proposed a deep one-stage neural network by employing a stacked convolutional encoder to generate their low-dimensional high-level representations aiming for maximal compactness. Furthermore, they incorporated a decoder to reconstruct raw samples from these low-dimensional representations. By employing proxy task learning, Liu et al. [19] introduced a spatial-temporal memory-augmented auto-encoder. Following this structure, researchers [25], [26], [27] developed the video prediction network that separately learns appearance and motion normality to predict the anomalies. Zhong et al. [28] utilized a cascade structure combining pixel reconstruction and optical flow prediction to enhance anomaly detection. Zeng et al. [29] used Graph Convolutional Neural Network to detect human-related anomalies. Jin et al. [23] utilized transformers that learn the discriminative temporal features among all video frames. Zhong et al. [30] developed an attention-based feature fusion module to integrate forward and backward spatio-temporal features, and utilizes an error pyramid and mean pooling for anomaly evaluation, effectively detecting objects of varying sizes in complex scenes. Li et al. [31] leveraged multi-branch generative adversarial network to detect abnormal events.

2) *Weakly Supervised VAD (WSVAD)*: Weakly supervised video anomaly detection methods have garnered increased attention, particularly following the introduction of the weakly supervised Multiple Instance Learning (MIL) framework by Sultani et al. [9]. They conceptualized each video as a bag, with its segments representing instances within the bag. The framework generates results based on the premise that the anomaly score for instances within the positive bag (anomalous videos) should surpass that of the negative bag (normal videos), utilizing ranking loss functions with video-level annotations. Therefore, several research works follow this direction which has attracted the attention of the research community. Zhong et al. [32] utilized a Graph Convolutional Neural Network to correct label noise in the area of video anomaly detection. Lv et al. [33] presented a localization framework by employing a high-level context information module for anomaly localization. Tian et al. [34] developed a robust temporal feature magnitude learning framework by leveraging a self-attention network and dilated convolutions to capture short and long-range temporal dependencies. Huang et al. [35] proposed a temporal feature aggregator for modeling the temporal relationships between segments of video. Moreover, they also incorporated a discriminative feature encoder for feature

discrimination. Liu et al. [36] obtained deep discriminative features by introducing a collaborative learning module. Chen et al. [37] applied a magnitude contrastive loss to capture feature separability between normal and abnormal videos effectively. Fan et al. [38] proposed a snippet-level attention mechanism to effectively localize the anomalous events in surveillance videos.

B. Vision-Language Model (VLM)

Vision-language models have garnered significant attention from the computer vision community due to their ability to bridge the gap between visual data and natural language. Earlier VLMs focused on image applications such as image classification [12], scene text detection [39], and image captioning [40]. Recently, Xu et al. [41] extended the VLM from image to video domains that aligns video and text by contrasting temporally related video-text pairs showing robust zero-shot capabilities in video tasks. Ju et al. [15] developed a simple baseline for efficient action recognition task by learning the task-specific prompt templates. Zanella et al. [42] leveraged the latent CLIP feature space to identify the normal events and establish text-driven vectors for detecting abnormalities. Wu et al. [16] adapted the frozen CLIP model to effectively transfer pre-trained language-visual knowledge to WS-VAD. Yang et al. [43] proposed a novel framework by utilizing the CLIP model to align video event description texts with corresponding video frames, improving pseudo-label generation and self-training through fine-tuning, learnable text prompts, and normality guidance. Wu et al. [44] introduced a new paradigm for open-vocabulary video anomaly detection using the CLIP model to improve detection and categorization of seen and unseen anomalies.

The following shortcomings of video anomaly detection methods based on the vision-language model are identified from the related works:

- Despite the effectiveness of existing approaches, they do not effectively capture the dynamic and temporal dependencies in videos. To address this challenge, we have introduced a Glimpse and Emphasize Network (detailed in Section III-C).
- Current state-of-the-art methods fail to fully exploit the connection between visual and language features, resulting in suboptimal performance. This limitation is addressed through a dual-block structure, where one branch focuses on visual features for coarse-grained binary classification, and the other integrates both visual and textual features for fine-grained anomaly detection (discussed in Section III-D and III-E).
- Existing methods rely on hand-crafted or learnable prompts, which are inadequate for capturing the subtle anomalies. To overcome this issue, we have incorporated a reparameterized learnable prompt (explained in Section III-F).

III. METHODOLOGY

This section provides a detailed description of the proposed ReFLIP-VAD, which addresses the shortcomings discussed in

Section II-B. Section III-A defines the problem formulation of weakly supervised anomaly detection in videos. Section III-B outlines the process for extracting both visual and textual features. Section III-C presents the glimpse and emphasize network of how global and local context information is captured effectively. Section III-D and III-E describe the classification and video-text alignment block respectively. Section III-F introduces the novel reparameterized learnable prompt using the DistilBERT encoder. Section III-G outlines how visual and text embeddings are combined. Finally, the coarse-grained and fine-grained alignment along with a novel feature magnitude separability loss functions are described. The overall framework of the proposed method illustrated in Figure 1.

A. Problem Definition

Consider a training set of untrimmed videos $\mathcal{U} = \{u_k\}_{k=1}^{N_{uv}}$, where N_{uv} denotes the total number of untrimmed videos. Each video u_k is associated with a binary label $l_k \in \{0, 1\}$ from the set $\mathcal{L} = \{l_k\}_{k=1}^{N_{uv}}$, where $l_k = 1$ indicates that if at least one frame of the video contains abnormal events and $l_k = 0$ otherwise. The WSVAD model aims to predict an anomaly score vector $\mathbf{sv}_k \in \mathbb{R}^{S_k}$ for each video, where S_k is the number of segments in the k -th video, and each element of $\mathbf{sv}_k$ represents a normalized anomaly score for each segment in the range from 0 to 1.

B. Feature Extraction

Previous research studies typically employ pre-trained 3D convolutional models like C3D and I3D to extract features from videos. These features are then input into MIL-based binary classifiers. Recently, FLIP [13] a large-scale language-vision pre-trained model, has significantly impacted various computer vision fields, showcasing strong generalization capabilities across diverse tasks. Drawing inspiration from FLIP, the proposed work not only utilizes the image encoder of FLIP to extract video features but also explores leveraging text of encoder of FLIP to fully utilize the rich associations between visual content and textual concepts.

For the visual features, the image encoder is a Vision Transformer that concatenates an additional [CLASS] token embedding and projects image patches linearly as input [45]. For the textual features, following [10], a Byte-Pair Encoding (BPE) [46] with a vocabulary size of 49,408 is used to tokenize the text. Each text sequence starts with a Beginning of Sentence [BOS] token and ends with an End of Sentence [EOS] token. The token embeddings are then fed into a modified decoder-only Transformer model [47]. Textual tokens and visual tokens are then projected into a multi-modal common space and separately L2-normalized. This approach differs from existing dual-stream models, which only model cross-modal interaction via global features. FLIP introduces a fine-grained contrastive learning objective that accounts for the interaction between image patches and textual tokens.

Initially, the FLIP image feature extractor processes a set of N_{uv} raw untrimmed videos $\mathcal{U}$. For each video $u_k \in \mathbb{R}^{N_f \times \mathcal{X} \times \mathcal{Y} \times 3}$, the parameters N_f , $\mathcal{X}$, and $\mathcal{Y}$ denote the number of frames, height, and width of u_k , respectively. Following

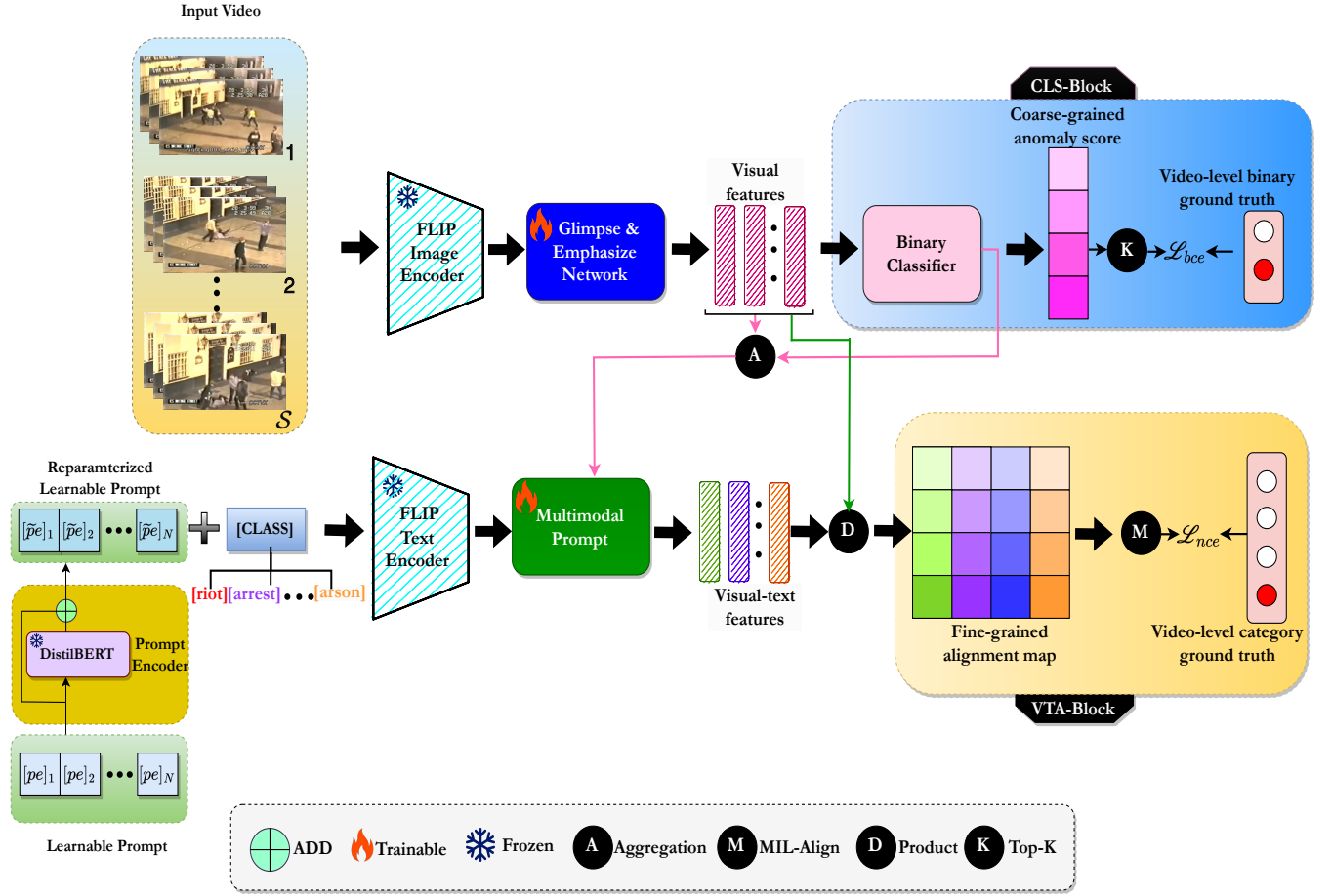


Fig. 1. Proposed Framework

this, each video sequence is partitioned into S segments and the resulting feature maps produced by the feature extractor are denoted by $\mathcal{G} = \{g^{k,j}\}$, where k varies from 1 to N_{uv} , and j varies from 1 to S . These feature maps are quantified as $\mathcal{G} \in \mathbb{R}^{N_{uv} \times S \times C \times D}$, with C indicating the number of crops in each video segment and D representing the dimensionality of features. Each $g^{k,j} \in \mathbb{R}^{C \times D}$ corresponds to the feature set of the j -th segment in the k -th video.

Furthermore, the Feature Intensify Module (FIM) module initially processes a feature map $\mathcal{G}$ to compute a feature norm $\mathcal{F}$ as will be discussed later. Subsequently, the Glimpse and Emphasize module leverages this enhanced feature map to extract global and local representation through a segment-level transformer (SLT) and self-attention mechanism (SAM), respectively. Initially, FIM computes the feature norm $\mathcal{F}^{k,j}$ of $g^{k,j}$ as per Eq. (1):

$$\mathcal{F}^{k,j} = \left(\sum_{d=1}^D |g^{k,j,d}|^2 \right)^{\frac{1}{2}} \in \mathbb{R}^{1 \times 1 \times C \times 1} \quad (1)$$

where d is the index of the feature dimension.

Subsequently, FIM enhances the features $\{g^{k,j}\}_{FIM}$ by integrating the modulated feature norm, $\text{Conv1D}(\mathcal{F}^{k,j})$, with $g^{k,j}$ acting as a residual component as illustrated in Eq. (2):

$$g_{FIM}^{k,j} = g^{k,j} + \alpha \text{Conv1D}(\mathcal{F}^{k,j}) \in \mathbb{R}^{1 \times 1 \times C \times D} \quad (2)$$

Here, α signifies a hyperparameter governing the influence of the norm term, and Conv1D is a 1-dimensional convolutional layer that modulates the feature norm for every dimension.

C. Glimpse and Emphasize Network

1) *Glimpse Module*: The glimpse module comprises three key components: channel squeezer (CS), segment level transformer (SLT), and feed-forward network (FFN). To streamline computation, the channel squeezer reduces the dimensionality of the feature map from D in the Feature Intensify Module (FIM) to $D/32$ that generates an output feature map $\mathcal{G}_{cs_global} \in \mathbb{R}^{U \times C \times S \times D/32}$. Furthermore, a video segment level transformer (SLT) is implemented to learn the global feature relationships across video segments. This is achieved by formulating an attention map $\mathcal{M} \in \mathbb{R}^{1 \times S \times S \times C}$ that distinctly associates the various temporal segments.

$$\mathcal{M}^{k,j_1,j_2} = \sum_{d=1}^D Q \left(\mathcal{G}_{cs_global}^{k,j_1,d} \right) K \left(\mathcal{G}_{cs_global}^{k,j_2,d} \right) \quad (3)$$

where, $j_1, j_2 \in [1, S]$, Q, K represent 1D “query” and “key” convolutions of the transformer. Then, softmax normalization is applied to obtain $m \in \mathbb{R}^{1 \times S \times S \times D}$, where $m^{k,j_1,:}$ indicates the association of other segments with the clip j_1 .

$$m^{k,j_1,j_2} = \frac{e^{\mathcal{M}^{k,j_1,j_2}}}{\sum_{j_2=1}^S e^{\mathcal{M}^{k,j_1,j_2}}} \quad (4)$$

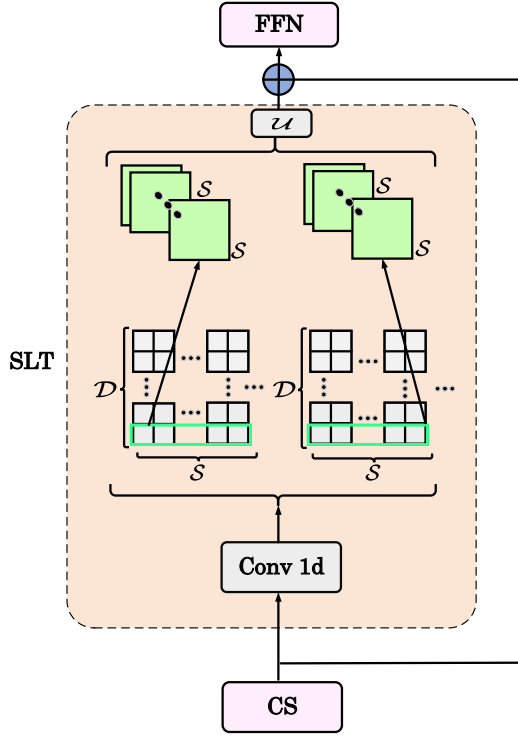


Fig. 2. Glimpse Module introduced in III-C to capture the global features.

The output $\mathcal{G}_{slt_global}$ of SLT is in $\mathbb{R}^{1 \times S \times C \times \frac{D}{32}}$, reflecting the weighted average of all segment features in the long video.

$$\mathcal{G}_{slt_global}^{k,j_1,d} = \sum_{j_2=1}^S m^{k,j_1,j_2} V \left(\mathcal{G}_{cs_global}^{k,j_2,d} \right) \quad (5)$$

Here, the Glimpse module within the network structure as shown in Figure 2 uses a 1D “value” convolution represented by V from the transformer architecture. This module equips the network with an understanding of typical, expected scenarios, which enhances its ability to pinpoint anomalies. Moreover, it leverages this insight to make better use of extensive temporal contexts within the data. The Glimpse module also incorporates a Feed-Forward Network (FFN), consisting of a pair of fully connected layers alongside a Gaussian Error Linear Unit (GeLU) non-linear function. This setup is designed to enhance the ability of the model to represent and process complex features. The resulting feature map from the Glimpse module is then directed into the subsequent Emphasize module for further analysis.

2) *Emphasize Module*: The emphasize module comprises three key components: channel squeezer (CS), self-attention mechanism (SAM), and the feed-forward network (FFN). The channel squeezer effectively reduces the feature map from its original channel dimension D to $D/16$. The self-attention mechanism (SAM) enhances the local features in each video segment by learning channel-wise correlations. It allows the model to focus on relevant information and capture context. The feed-forward network contributes to the generation of a refined feature map. This refined map serves as an indicator of anomalies present in the video segments.

The feature map $\mathcal{G}_{sam_local}$ is computed by the self-attentional

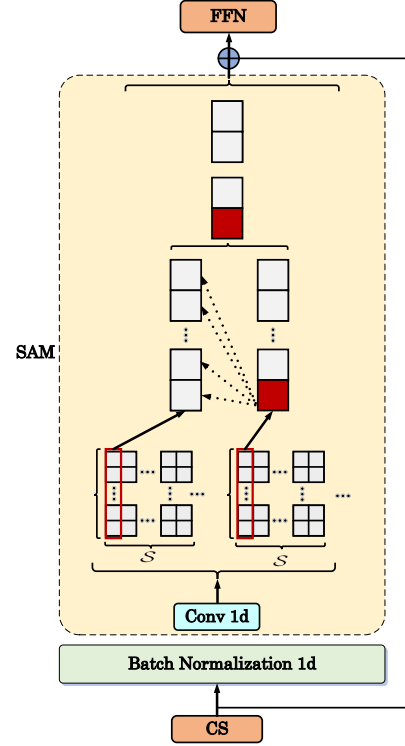


Fig. 3. Emphasize module introduced in III-C to capture the local features.

convolution on $\mathcal{G}_{ca_local}$ which is represented in Eq. (5). The operation is represented by a Hadamard product (denoted by $\otimes$) of $\mathcal{G}_{ca_local}$ with itself, resulting in a tensor of dimensions $\mathcal{U} \times S \times C \times \frac{D}{16}$. The individual elements of $\mathcal{G}_{sam_local}$ are computed by summing over the $\frac{D}{16}$ channels of the Hadamard product of the corresponding elements of $\mathcal{G}_{ca_local}$ represented in Eq. (6), without the use of learnable weights. This process facilitates each channel in accessing adjacent channels to learn channel-wise correlations. Following this operation, a two-layered feedforward network (FFN) processes the output to produce the final feature map as shown in Figure 3.

$$\mathcal{G}_{sam_local} = \mathcal{G}_{ca_local} \otimes \mathcal{G}_{ca_local} \in \mathbb{R}^{\mathcal{U} \times S \times C \times \frac{D}{16}} \quad (6)$$

where for each element of $\mathcal{G}_{sam_local}$, we have:

$$\mathcal{G}_{sam_local}^{k,j,l_1} = \sum_{l_2=0}^{\frac{D}{16}} \mathcal{G}_{ca_local}^{k,j,l_1} \cdot \mathcal{G}_{ca_local}^{k,j,l_2} \in \mathbb{R}^{1 \times 1 \times C \times 1} \quad (7)$$

D. Classification Block (CLS Block)

Unlike previous WSVAD approaches, ReFLIP-VAD incorporates a two-pronged block architecture. Following temporal modeling, the local video features $\mathcal{G}_{sam_local}$ are passed through a fully connected (FC) layer to produce the final visual features as $\mathcal{G}_{visual}$. Typically, in the CLS block, $\mathcal{G}_{visual}$ is fed into a binary classifier. This classifier includes a feed-forward network layer (FFN), a fully connected layer (FN),

and a Sigmoid activation(σ) to calculate the coarse-grained anomaly confidence score $\mathcal{A}$, given in Eq. (8):

$$\mathcal{A} = \sigma(\text{FC}(\text{FFN}(\mathcal{G}_{\text{visual}}) + \mathcal{G}_{\text{visual}})) \quad (8)$$

E. Video-Text Alignment Block (VTA-Block)

In addition to the conventional binary classification block, a novel video-text alignment block is introduced. In this block, textual labels such as ‘explosion’, ‘road accident’, ‘arson’, ‘abuse’ etc., are not encoded as one-hot vectors; rather, they are converted into class embeddings using the text encoder of FLIP. By leveraging the static pre-trained text encoder of FLIP, we harness its linguistic knowledge for enhancing video anomaly detection. Subsequently, the similarities between class embeddings and frame-level visual features are assessed to generate the alignment map $\mathcal{AM} \in \mathbb{R}^{n \times \ell}$, where ℓ is the number of textual labels. This arrangement mirrors the setup in CLIP, where each input text label corresponds to a distinct class of abnormal events, facilitating fine-grained WS-VAD.

F. Reparameterized Learnable Prompt

The proposed method is built upon the CLIP [10], a visual-language model consisting of the visual encoder and text encoder that transforms images and text into visual and textual embeddings. CLIP generates the textual label embeddings by leveraging manual prompts. However, such handcrafted prompts pose a significant challenge because it is highly time-consuming. To address this issue, a novel approach is introduced: **Reparameterized Fine-grained Language-Image Pre-training (ReFLIP)**. Instead of directly optimizing prompts, ReFLIP employs a reparameterized prompt encoder (DistilBERT) as depicted in Figure 4 to re-parameterize input prompt embeddings. This reparameterization enhances the exploration of task-specific knowledge from few-shot samples.

Given a set of original prompt embeddings PE consisting of N prompt tokens $PE = \{pe_1, pe_2, \dots, pe_N\}$, the prompt encoder is applied to obtain a reparameterized sequence $\widetilde{PE}$ as follows:

$$\widetilde{PE} = \{\widetilde{pe}_1, \widetilde{pe}_2, \dots, \widetilde{pe}_N\} = \{\phi(pe_1), \phi(pe_2), \dots, \phi(pe_N)\}$$

where $\phi(\cdot)$ denotes the reparameterization function of the prompt encoder, which includes a network $\theta(\cdot)$ with a residual connection. The function ϕ is defined for each prompt token pe_i as:

$$\phi(pe_i) = \theta(pe_i) + pe_i, \quad i \in \{1 \dots N\}$$

The network $\theta(\cdot)$ is an adaptable approach that forms connections between the prompt embeddings. This facilitates the creation of task-specific reparameterization within the prompt embeddings. After applying reparameterization, the class token embedding c_i for the i -th class is integrated into the center of the reparameterized learnable prompts $\phi(pe_i)$ forming the sequence as $p_i = \{\phi(pe_1), \phi(pe_2), \dots, c_i, \dots, \phi(pe_N)\}$. Following their integration with positional embeddings to incorporate positional information, these prompts are fed into the FLIP textual encoder $e_{\text{text}}(\cdot)$. This process generates the textual class embeddings, represented as $\mathcal{T}_i = e_{\text{text}}(p_i)$.

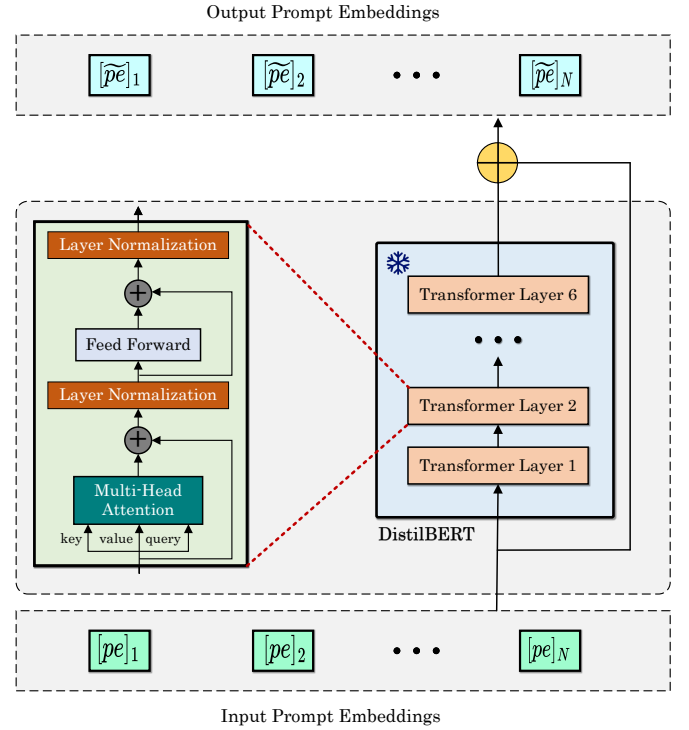


Fig. 4. Reparameterized Learnable Prompt using DistilBERT Architecture

G. Multimodal Prompt

To enhance text label representations for abnormal events, we introduce a method utilizing visual contexts to refine class embeddings. This involves an anomaly-centric visual prompt leveraging the visual embeddings from abnormal video segments to improve text label accuracy. The anomaly confidence vector from the Classification block serves as anomaly attention. The video-level prompt is calculated by the dot(.) product of this anomaly attention with the video feature vector, followed by normalization. The visual prompt is then integrated with the class embedding, and the final segment-specific class embedding is generated using a Feedforward Neural Network with an addition operation and a skip connection.

$$\mathcal{V} = \text{Norm}(\mathcal{A}^T \cdot \mathcal{G}_{\text{visual}}) \quad (9)$$

where $\mathcal{V} \in \mathbb{R}^d$ is the anomaly-centric visual prompt. We then add $\mathcal{V}$ to the class embedding $\mathcal{T}$ and obtain the final instance-specific class embedding $\mathcal{CE}$ by a simple FFN layer and a skip connection.

$$\mathcal{CE} = \text{FFN}(\text{ADD}(\mathcal{V}, \mathcal{T})) + \mathcal{T} \quad (10)$$

where ADD denotes element-wise addition. The fine-grained alignment map $\mathcal{AM}$ is then generated by calculating the similarity scores across all instance-specific class embeddings and frame-level visual features.

H. Loss function

For the CLS block, the WSVAD task handles known anomalies using either MIL or Top- $\mathcal{K}$ due to the lack of precise frame-level annotations. It utilizes the Top- $\mathcal{K}$ mechanism to

identify $\mathcal{K}$ highest anomaly confidences. It calculates the classification loss using binary cross entropy between the video-level predictions and the actual ground truth. In contrast, the Video-Text Alignment-block (VTA-block) addresses scenarios where there are no anomaly confidences and the challenge of dealing with multiple classes instead of binary classes. Taking inspiration from Wu et al. [16], the MIL-Align mechanism is utilized to address the above challenge. The MIL-Align mechanism focuses on the alignment map $\mathcal{AM}$, which captures the similarity between frame-level video features and all class embeddings. By selecting the top $\mathcal{K}$ similarities and averaging them, the degree of fine-grained alignment for a given video with the current class is assessed such that it exhibits the highest similarity score among all options. The probability for multi-class classification is computed as the softmax normalized similarity between visual embeddings $\mathcal{V}$ and all text embeddings $\mathcal{T}$ for each category.

$$p_i^{vt} = \frac{\exp(\text{sim}(\mathcal{V}, \mathcal{T}_i)/\tau)}{\sum_{j=1}^{K+1} \exp(\text{sim}(\mathcal{V}, \mathcal{T}_j)/\tau)} \quad (11)$$

where p_i^{vt} denotes the predicted probability corresponding to the i -th category, $\text{sim}(\cdot, \cdot)$ is the cosine similarity and $K + 1$ is the K number of anomaly categories and 1 normal category. τ represents the temperature scaling parameter that adjusts the softmax distribution. The fine-grained alignment loss $\mathcal{L}_{fga}$ is calculated using the Kullback-Leibler divergence. This approach compels the network to discern between the visual content of the video that depicts abnormal behavior (foreground) and the content irrelevant to the abnormal behavior (background).

$$\mathcal{L}_{fga} = \mathbb{E}_{p \sim p(v)} [\log p^{vt}(v) - \log q^{vt}(v)] \quad (12)$$

where $p^{vt}(v)$ and $q^{vt}(v)$ denote the similarity score and semantic consistency label of the visual-prompt pair, respectively. If it is a positive pair, $q = 1$; otherwise, $q = 0$. The coarse-grained alignment loss is binary cross entropy loss $\mathcal{L}_{cga}$ between p_i and l_k as given in Eq. (13).

$$\mathcal{L}_{cga} = -\frac{1}{N} \sum_{i=1}^N [l_k \log(p_i) + (1 - l_k) \log(1 - p_i)] \quad (13)$$

where, N is the number of video segments, l_k is the ground-truth label for the i -th video segment, where $l_k = 1$ if the segment is anomalous and $l_k = 0$ if it is normal. p_i is the predicted probability that the i -th video segment is anomalous.

This model incorporates a feature magnitude separability loss $\mathcal{L}_{fms}$ given in Eq. (14) in addition to the coarse-grained alignment loss $\mathcal{L}_{cga}$ and fine-grained alignment loss $\mathcal{L}_{fga}$. This separability loss serves to incrementally separate the normal class embeddings from the abnormal ones. The process begins by computing the cosine similarity between the normal and abnormal class embeddings. Subsequently, the contrastive loss $\mathcal{L}_{fms}$ is calculated as follows:

$$\begin{aligned} \mathcal{L}_{fms} = & \sum_{p,q=0}^{U/2} (1 - \mathcal{I})(\delta(\mathcal{W}_{normal}^p, \mathcal{W}_{normal}^q)) \\ & + \sum_{r,s=U/2}^U (1 - \mathcal{I})(\delta(\mathcal{W}_{abnormal}^r, \mathcal{W}_{abnormal}^s)) \\ & + \sum_{p=0}^{U/2} \sum_{r=U/2}^U \mathcal{I}(\text{Margin} - \delta(\mathcal{W}_{normal}^p, \mathcal{W}_{abnormal}^r)) \end{aligned} \quad (14)$$

Here, $U/2$ normal videos and $U/2$ abnormal videos are sampled in each training batch U . p, q represent the normal segments index, and r, s denote the abnormal segments index. $\mathcal{W}_{normal}$ and $\mathcal{W}_{abnormal}$ are the top- $\mathcal{K}$ feature magnitudes for normal and abnormal segments respectively. $\delta(\cdot, \cdot)$ is a distance metric used to increase the feature magnitude distance between the segments.

Here, $\mathcal{I} = 1$ indicates that the two sampled video segments (one normal and one abnormal) are of different categories. The loss function is to increase the distance between the feature magnitudes of these segments. The goal is to enhance the model's ability to distinguish clearly between normal and abnormal segments, which aids in better anomaly detection. $\mathcal{I} = 0$ indicates that the paired segments sampled are both from the same category, either both normal or both abnormal. In this scenario, the $\mathcal{L}_{fms}$ loss function aims to minimize the distance between the feature magnitudes of the segments, effectively grouping similar types of segments closer together in the feature space. This reinforces the model's capability to recognize and maintain consistency within the same categories of segments, thereby improving the overall model performance in distinguishing between normal and abnormal segments.

The distance functions $\delta(\mathcal{W}_{normal}^p, \mathcal{W}_{normal}^q)$ and $\delta(\mathcal{W}_{abnormal}^r, \mathcal{W}_{abnormal}^s)$ are defined in Eq. (15) and Eq. (16) respectively.

$$\delta(\mathcal{W}_{normal}^p, \mathcal{W}_{normal}^q) = \min_{u,v \in \{0, \dots, S\}} (\mathcal{O}(\|f_{EM}^{p,u}\|_2) - \mathcal{O}(\|f_{EM}^{q,v}\|_2)) \quad (15)$$

$$\delta(\mathcal{W}_{abnormal}^r, \mathcal{W}_{abnormal}^s) = \min_{u,v \in \{0, \dots, S\}} (\mathcal{O}(\|f_{EM}^{r,u}\|_2) - \mathcal{O}(\|f_{EM}^{s,v}\|_2)) \quad (16)$$

where

$\mathcal{O}(\|f_{EM}^{r,u}\|_2)$ is one of the top- $\mathcal{K}$ feature magnitudes among $f_{EM}^{r,\{1, \dots, S\}}$.

Similarly, the distance $\delta(\mathcal{W}_{normal}^p, \mathcal{W}_{abnormal}^r)$ for normal feature p and abnormal features r is presented in Eq. (17) as:

$$\delta(\mathcal{W}_{normal}^p, \mathcal{W}_{abnormal}^r) = \max_{u,v \in \{0, \dots, S\}} (\mathcal{O}(\|f_{EM}^{p,u}\|_2) - \mathcal{O}(\|f_{EM}^{r,v}\|_2)) \quad (17)$$

Here, the max operation is taken over all temporal segments u and v indicate the largest difference in feature magnitudes between the normal and abnormal features.

To sum up, the overall loss for ReFLIP-VAD is expressed in Eq. (18) as:

$$\mathcal{L} = \mathcal{L}_{cga} + \lambda_1 \mathcal{L}_{fga} + \lambda_2 \mathcal{L}_{fms} \quad (18)$$

where λ_1, λ_2 is the weighting factor that adjusts the influence of loss terms.

IV. EXPERIMENTS

A. Datasets and Evaluation Metrics

1) *Datasets*: Extensive experiments have been conducted on two standard benchmark datasets, namely UCF-Crime [9] and XD-Violence [48].

UCF-Crime is a large-scale weakly supervised video anomaly detection dataset released in 2018. It comprises 1900 untrimmed real-world videos with a total duration of 128 hours. It includes 13 different classes of anomalous events. It is split into a training set of 1610 videos and a testing set of 290 videos.

XD-violence is a multi-modal large-scale diverse dataset released in 2020. It is collected from movies, surveillance cameras, and in-the-wild scenes for 217 hours. It includes 6 different classes of physically violent events. It comprises 4754 videos, with 2,349 categorized as normal and 2,405 as abnormal. The training set comprises 3,954 videos, while the test set contains 800 videos.

2) *Evaluation Metrics*: For evaluating coarse-grained Weakly Supervised Video Anomaly Detection (WSVAD), this study adopts the frame-level Average Precision (AP) metric, which calculates the area under the precision-recall curve for the XD-Violence dataset, and the frame-level Area Under the Curve (AUC) for the UCF-Crime dataset. Additionally, the AUC for only the anomalous videos (referred to as AnoAUC) is utilized to highlight the detection accuracy within anomalous content, addressing the issue where high performance on normal videos might obscure lower detection accuracy within abnormal videos. For fine-grained WSVAD, the study follows the established protocol in video action detection, employing the mean Average Precision (mAP) across varying Intersection over Union (IoU) thresholds. Specifically, IoU thresholds from 0.1 to 0.5, in increments of 0.1, are used to calculate mAP values. The average of these mAP values (AVG) is also reported. It is important to note that mAP calculations are performed solely on the anomalous videos within the test set.

B. Implementation Details

The model is trained on NVIDIA RTX 4070 GPU using the PyTorch framework. For the optimization of the network, Adam is leveraged as the optimizer with a batch size of 128. On the XD-Violence dataset, the learning rate and total epoch are set as 5×10^{-4} and 50, respectively, and on the UCF-Crime dataset, the learning rate and total epoch are set as 3×10^{-4} and 50, respectively. For hyper-parameters, τ in Eq. (11) is set as 0.07. For the loss weight adjustment factor in Eq. (18), the values of λ_1 and λ_2 are configured as 0.01 and 0.001 on the UCF-Crime dataset, respectively. On the XD-Violence dataset, both λ_1 and λ_2 are set as 0.0001. For the MIL-ranking, the value of $\mathcal{K}$ as $\lfloor \frac{S}{16} + 1 \rfloor$ for positive bags and

$\mathcal{K}$ as 1 for negative bags, where S denotes the length of input feature. To make a fair comparison, the visual sample rate is set to be 24 fps for UCF-Crime and 30 fps for XD-violence as prior methods [14] and a sliding window with a size of 16 frames for visual feature extraction.

C. Comparison with State-of-the-art Methods

1) *Coarse-grained WSVAD Results*: The proposed method is designed to handle both coarse-grained and fine-grained WSVAD tasks effectively. To demonstrate its performance, the method is compared with other state-of-the-art methods in Tables I and II. This approach ensures that the comparisons are equitable and highlights the capabilities of the proposed method in anomaly detection.

TABLE I
PERFORMANCE COMPARISON(COARSE-GRAINED) ON UCF-CRIME

Supervision	Method	Features	AUC(%)	Ano-AUC(%)
Un	SVM baseline	-	50.10	50.00
	OCSVM [49]	-	63.20	51.06
	Hasan et al. [50]	-	51.20	39.43
Weak	Deep-MIL [9]	C3D	75.42	54.25
	IBL [51]	C3D	78.66	-
	Motion-Aware [52]	PWC Flow	79.01	62.18
	GCN [32]	TSN	82.12	59.03
	HL-Net [14]	I3D	82.45	60.27
	MS-BSAD [53]	I3D	83.54	-
	Ju et al. [15]	CLIP	84.72	62.60
	CRFD [54]	I3D	84.89	-
	LA-Net [55]	I3D	85.12	-
	NG-MIL [56]	I3D	85.63	-
	RTFM [34]	I3D	85.66	63.86
	AnomalyCLIP [42]	ViT-B/16	86.36	-
	DMU [57]	I3D	86.75	66.8
	UB-MIL [58]	X-CLIP	86.97	68.94
	CLIP-TSA [59]	CLIP	87.58	69.31
	Vad-CLIP [16]	CLIP	88.02	70.23
Weak	ReFLIP-VAD (Ours)	CLIP	88.57	72.35
	ReFLIP-VAD (Ours)	FLIP	89.14	74.72

Table I presents a performance comparison of existing methods on the UCF-Crime dataset, emphasizing both coarse-grained AUC and anomaly-specific detection capabilities (Ano-AUC). The proposed method achieves an impressive AUC of 89.14% and an Ano-AUC of 74.72%, showing substantial improvements over existing methods. Specifically, when compared to the highest scores from previous methods, VadCLIP [16] which scored 88.02% in AUC and 70.23% in Ano-AUC, the proposed method demonstrates an improvement of +1.12% in AUC and +4.49% in Ano-AUC. Additionally, compared to the second highest-performing method, CLIP-TSA [59], which scored an AUC of 87.58% and an Ano-AUC of 69.31%, the proposed method exhibits improvements of +1.56% in AUC and +5.41% in Ano-AUC. These enhancements significantly underscore the efficacy of the proposed method in distinguishing between normal and abnormal events, setting a new benchmark for anomaly detection in the UCF-Crime dataset.

Table II showcases a performance comparison of existing anomaly detection methods on the XD-Violence dataset, categorized under unsupervised and weakly supervised techniques.

TABLE II
PERFORMANCE COMPARISON (COARSE-GRAINED) ON XD-VIOLENCE

Supervision	Method	Features	AP(%)
Un	Baseline	-	50.80
	OCSVM [49]	-	28.63
	Hasan et al. [50]	-	31.25
Weak	Deep-MIL [9]	C3D	75.18
	CRFD [54]	I3D	75.90
	Ju et al. [15]	I3D	76.57
	ITFM [34]	I3D	78.27
	HL-Net [14]	I3D	78.10
	NG-MIL [56]	I3D	78.51
	AnomalyCLIP [42]	ViT-B/16	78.55
	MS-BSAD [53]	I3D	78.92
	LA-Net [55]	I3D	80.72
	CMRL [60]	I3D	81.30
	DMU [57]	I3D	82.41
	CLIP-TSA [59]	CLIP	82.17
Weak	Vad-CLIP [16]	CLIP	84.51
	ReFLIP-VAD (Ours)	CLIP	85.81
	ReFLIP-VAD (Ours)	FLIP	86.29

TABLE III
PERFORMANCE COMPARISONS(FINE-GRAINED) ON UCF-CRIME

Method	mAP@IoU(%)					
	0.1	0.2	0.3	0.4	0.5	AVG
Baseline	0.21	0.14	0.04	0.02	0.01	0.08
Deep-MIL [9]	5.73	4.41	2.69	1.93	1.44	3.24
HL-Net [14]	10.27	7.01	6.25	3.42	3.29	6.05
Vad-CLIP [16]	11.72	7.83	6.40	4.53	2.93	6.68
ReFLIP-VAD (Ours)	14.23	10.34	9.32	7.54	6.81	9.62

TABLE IV
PERFORMANCE COMPARISONS(FINE-GRAINED) ON XD-VIOLENCE

Method	mAP@IOU(%)					
	0.1	0.2	0.3	0.4	0.5	AVG
Baseline	1.82	0.92	0.48	0.23	0.09	0.71
Deep-MIL [9]	22.72	15.57	9.98	6.20	3.78	11.65
HL-Net [14]	30.51	25.75	20.18	14.83	9.79	20.21
Vad-CLIP [16]	37.03	30.84	23.38	17.90	14.31	24.70
ReFLIP-VAD (Ours)	39.24	33.45	27.71	20.86	17.22	27.36

The Proposed Method achieves a notable Average Precision (AP) of 86.29%, which is substantially higher compared to both categories of existing methods. Among unsupervised methods, the Baseline method scores 50.80%, OCSVM [49] achieves 28.63%, and Hasan et al. [50] achieve 31.25%. Compared to these, the proposed method shows improvements of 35.49%, 57.66%, and 54.04% respectively. In the realm of weakly supervised methods, the highest previous AP was achieved by VadCLIP [16] at 84.51% and the lowest by Deep-MIL [9] at 75.18%. The proposed method surpasses UB-MIL [58] the second best-performing existing method in this category, by 4.12%. This significant outperformance of the proposed method over both unsupervised and weakly supervised methods underscores its superior efficacy in accurately identifying anomalies in the XD-Violence dataset.

2) *Fine-grained WSVAD Results:* For the fine-grained performance on two benchmark dataset, ReFLIP-VAD is compared with previous works [9], [14], [16] in Tables III and IV. HL-Net [14], the first work to propose fine-grained WSVAD, is

re-implemented using visual features from CLIP, followed by the setup in HL-Net to fine-tune Deep-MIL [9] for adapting to fine-grained WSVAD. Fine-grained WSVAD presents a more challenging task relative to coarse-fined WSVAD as it requires attention to both multi-category classification accuracy and detection segment continuity. In this task, the proposed method is also clearly superior to these excellent comparison methods on both XD-Violence and UCF-Crime datasets. For example, on XD-Violence, the ReFLIP-VAD achieves performance improvements of 15.71% , 7.15% and 2.66% in terms of AVG compared to Deep-MIL [9], HL-Net [14] and VadCLIP [16], respectively.

D. Qualitative Analysis

Figures 5 and 6 illustrate the qualitative visualization results for the UCF Crime and XD Violence datasets, respectively. The frame-wise anomaly detection scores are depicted by the blue solid line, and the ground-truth abnormal temporal locations are depicted by the light orange shaded regions. The proposed method has demonstrated effectiveness across various categories within both the UCF-Crime and XD-Violence datasets, effectively identifying anomalies within the designated ground truth regions. This confirms the model's precision in temporal and situational anomaly detection.

Figure 7 presents t-SNE visualizations of feature distributions on these datasets, both with and without the proposed loss function $\mathcal{L}$. The visualizations highlight that incorporating $\mathcal{L}$ significantly enhances the separation between normal (green dots) and abnormal (magenta dots) features, thereby improving the proposed model's discriminative capability. This improved distinction is crucial for effective anomaly detection, demonstrating the model's vital role in enhancing surveillance and safety systems by efficiently distinguishing between normal and anomalous activities.

E. Ablation Studies

Comprehensive ablation studies have been conducted to verify the effectiveness of the proposed model. For these studies, the similarity map is utilized to calculate the frame-level anomaly degree for coarse-grained WSVAD.

1) *Effectiveness of Temporal Modeling module:* To evaluate the efficacy of the temporal modeling module in the proposed framework, experiments are conducted on XD-Violence dataset in various settings.

TABLE V
EFFECTIVENESS OF TEMPORAL MODELING MODULE IN PROPOSED FRAMEWORK

Configuration	Method	AP(%)	AVG(%)
I	Baseline (w/o temporal modeling)	73.31	16.27
II	Global TF-Encoder	82.69	16.98
III	Emphasize-Glimpse (EG)	82.89	17.43
IV	Local TF-Encoder	81.34	18.67
V	Emphasize-Emphasize (EE)	81.59	19.27
VI	Graph Convolutional Network (GCN)	82.65	24.19
VII	Local TF-Encoder + Global TF-Encoder	80.12	20.93
VIII	Global TF-Encoder + GCN	85.15	21.76
IX	Local TF-Encoder + GCN	85.57	25.42
X	Glimpse-Emphasize-Fusion (GE-Fusion)	85.89	26.58
XI	Glimpse-Emphasize (GE) (Proposed Method)	86.29	27.36

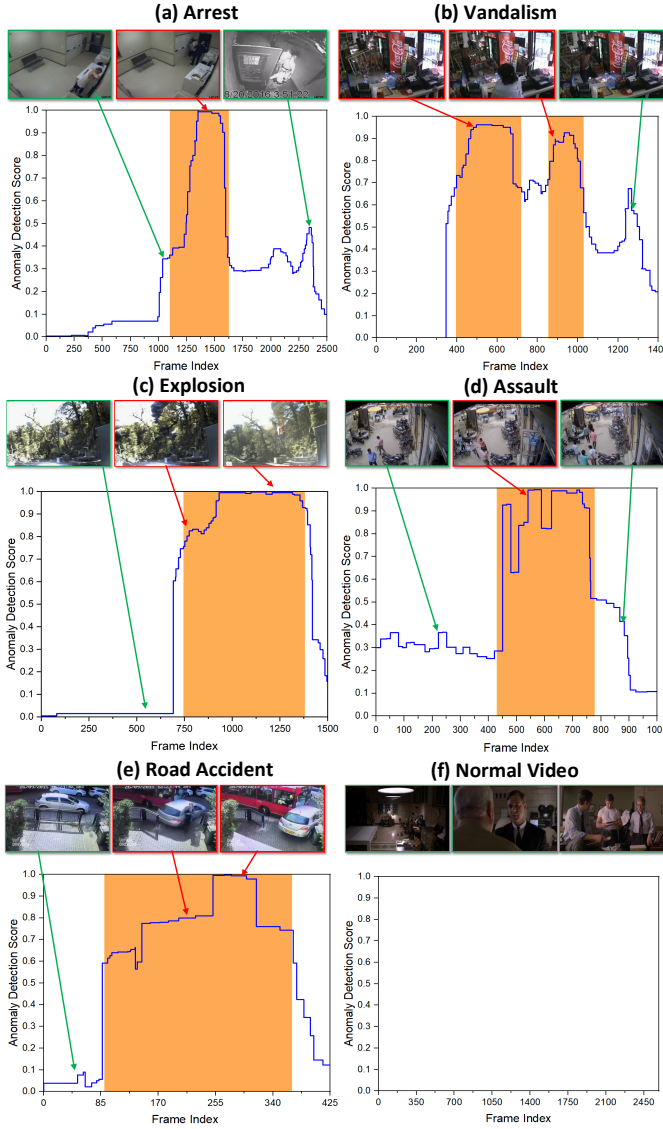


Fig. 5. Visualization Results of ground-truth and anomaly detection score of the proposed method on UCF-Crime dataset. The blue solid line depicts the frame-wise anomaly detection score and the light orange shaded regions indicate the ground-truth abnormal temporal locations. Green arrow represents the location of normal frame and red arrow represents the location of abnormal frame.

Table V illustrates the effectiveness of the Temporal Modeling module in the proposed framework across various configurations. Configuration I serves as the baseline without the temporal modeling capabilities of the proposed method, achieving an AP of 73.31% and an AVG of 16.27%. In Configuration II, the inclusion of the Global TF-Encoder enhances the AP and AVG scores to 82.69% and 16.98%, respectively. The Emphasize-Glimpse (EG) sequence in Configuration III further improves the scores slightly to 82.89% for AP and 17.43% for AVG. Compared to the baseline (Setting I), this configuration improves AP by +9.58% and AVG by +1.16%.

In Configurations IV and V, the Local TF-Encoder and the Emphasize-Emphasize (EE) arrangements show further improvements, with Configuration V (EE) achieving 81.59% AP and 19.27% AVG. This setting shows a decrease of -

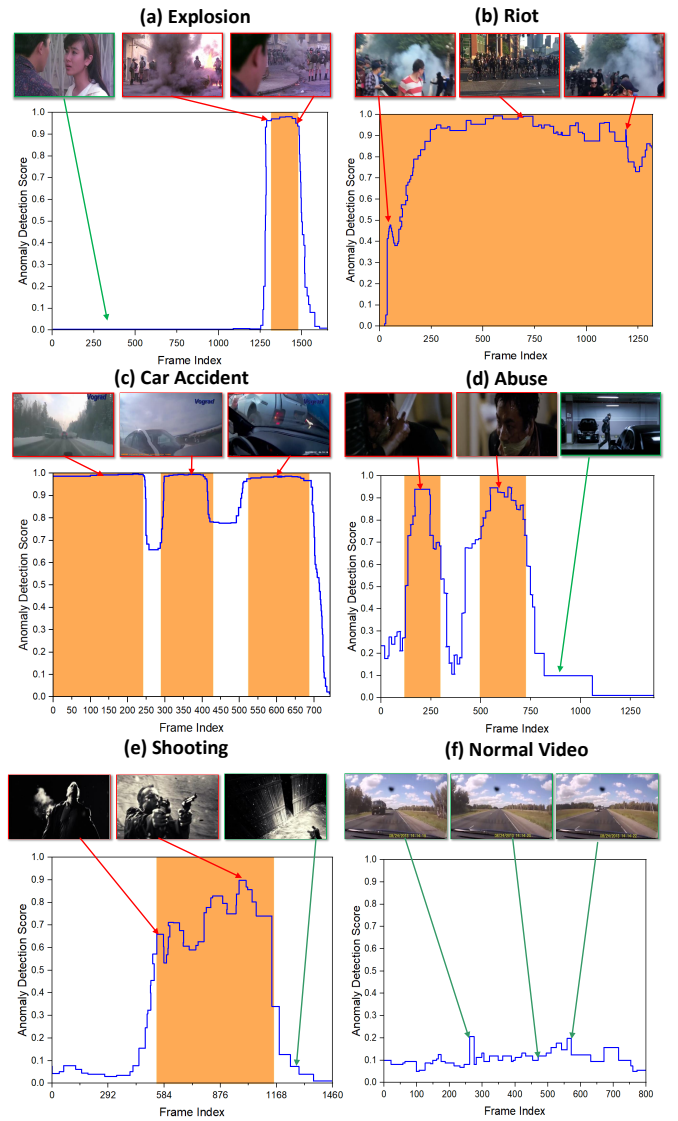


Fig. 6. Visualization Results of ground-truth and anomaly detection score of the proposed method on XD-Violence dataset. The blue solid line depicts the frame-wise anomaly detection score and the light orange shaded regions indicate the ground-truth abnormal temporal locations. Green arrow represents the location of normal frame and red arrow represents the location of abnormal frame.

1.35% in AP and -2.15% in AVG compared to Setting III. The integration of Graph Convolutional Network (GCN) in Configuration VI leads to an improvement, reaching 82.65% AP and 24.19% AVG. Configuration X shows an increase of +5.77% in AP and +5.65% in AVG compared to Configuration VII, an increase of +0.74% in AP and +4.82% in AVG compared to Configuration VIII, and an increase of +0.32% in AP and +1.16% in AVG compared to Configuration IX. However, it shows a notable decrease of -0.40% in AP and -0.78% in AVG compared to the proposed method in Setting XI. Configuration XI, representing the Glimpse-Emphasize (GE) method proposed in this paper, outperforms all other configurations, achieving the highest AP of 86.29% and an AVG of 27.36%. This emphasizes the efficacy of the GE approach, which leverages a sequential arrangement of the

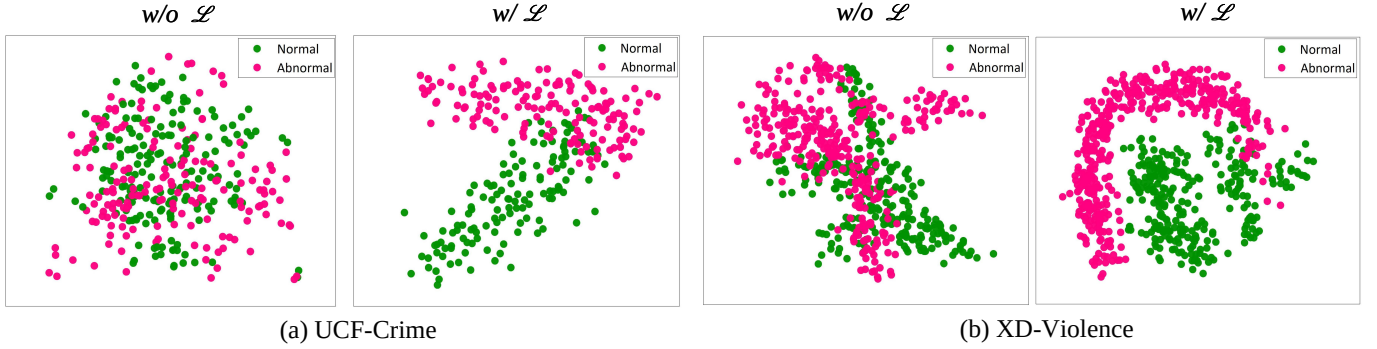


Fig. 7. Visualizations of the distributions of the normal and abnormal embeddings learned without and with the proposed loss ($\mathcal{L}$) using t-Distributed Stochastic Neighbor Embedding (t-SNE) [61], on (a) UCF-Crime dataset, and (b) XD-Violence dataset.

Glimpse and Emphasize modules, highlighting its advantage in handling temporal dependencies in video anomaly detection tasks.

2) *Effectiveness of dual block*: ReFLIP-VAD can simultaneously realize coarse-grained and fine-grained WSVAD tasks through a dual-block approach. To evaluate the effectiveness of each block and the prompt mechanisms in the video-text alignment (VTA-Block), extensive experiments were conducted, with detailed results presented in Table VI. The method utilizing only the Classification block (CLS-Block) falls under the classification-based paradigm and competes with current state-of-the-art methods on XD-Violence. However, using only the VTA block results in unsatisfactory performance in terms of AP, as it primarily focuses on fine-grained WSVAD. With the integration of coarse-grained classification for feature optimization in the CLS block, the VTA block experiences a significant improvement, approximately a 12% increase in AP. Furthermore, the dual block that integrates the Reparameterized Learnable Prompt (RL-prompt) and the Visual Prompt (V-prompt) to form a multimodal prompt achieves substantial improvements of +2.42% and +6.23%, respectively. Additionally, the incorporation of learnable prompts and visual prompts, which are specific designs in the VTA block, consistently enhances performance, establishing a new state-of-the-art. These results demonstrate that the dual-block configuration, which combines a coarse-grained classification paradigm with a fine-grained alignment paradigm, effectively increases the performance.

TABLE VI
EFFECTIVENESS OF DUAL BLOCK ON XD-VIOLENCE DATASET

CLS-Block	VTA-Block	RL-Prompt	V-Prompt	AP(%)
✓	✗	✗	✗	81.28
✗	✓	✗	✗	69.24
✓	✓	✗	✗	77.33
✓	✓	✓	✗	79.75
✓	✓	✗	✓	83.56
✓	✓	✓	✓	86.29

3) *Effectiveness of Context Length, Window Length, and Video Input Length*: As shown in Table VII, the performance metrics, AP(%) and AVG(%), tend to improve as the context length in the learnable prompt increases from 8 to 24. Specifically, the AP(%) reaches its peak at a context length of 24, yielding 86.29%, and an AVG(%) of 27.36%,

representing a substantial improvement of approximately 5% in AP and 4.62% in AVG compared to a context length of 8. However, further increasing the context length to 32 leads to a slight decrease in performance, with AP(%) dropping by 2.16% and AVG(%) by 4.47%, indicating that while a larger context length generally enhances the model's ability to capture relevant features, there is a threshold beyond which the benefits diminish.

The performance of the model is then analyzed by varying the window length of the Glimpse and Emphasize network. The optimal window length is identified between 32 and 64, where the AP(%) and AVG(%) reach their maximum values of 86.29% and 27.36%, respectively, showing an improvement of 2.11% in AP and 2.19% in AVG over the shortest window length of 16. However, extending the window length to 256 results in a notable decline in performance, with AP(%) decreasing by 2.75% and AVG(%) by 3.82%, indicating that excessively long window lengths might introduce unnecessary noise or irrelevant information, thereby affecting the model's robustness.

The input length directly influences the amount of video data processed by the model. As input length increases from 64 to 512, both AP(%) and AVG(%) improve, peaking at an input length of 512 with a substantial improvement of 4.69% in AP and 4.49% in AVG compared to the shortest input length. This demonstrates that the method is capable of effectively handling larger input sizes, leading to enhanced performance.

4) *Scale of Vision-Language Models (VL-Models) and Their Impact on Fine-Grained WSVAD*: Table VIII illustrates the performance impact of different Vision-Language Model (VL-Model) variants on fine-grained WSVAD task. The DistilBERT + FLIP model achieves the highest mAP@IoU=0.3 of 27.71% and an AUC of 86.29%, representing a significant improvement over the base CLIP model, which records an mAP of 22.38% and an AUC of 81.19%. Specifically, the DistilBERT + FLIP variant outperforms the CLIP model by 23.82% in mAP and 6.27% in AUC. Compared to ALIGN, which has the largest model size at 307M parameters, DistilBERT + FLIP still shows a 9.37% increase in mAP and a 1.77% improvement in AUC, despite having a considerably smaller model size of 165M parameters. This highlights the effectiveness and efficiency of integrating DistilBERT with FLIP for fine-grained WSVAD tasks, offering a better trade-off between model size

TABLE VII
IMPACT OF CONTEXT LENGTH, WINDOW LENGTH, AND VIDEO INPUT LENGTH

Context Length	AP(%)	AVG(%)	Window Length	AP(%)	AVG(%)	Input Length	AP(%)	AVG(%)
8	81.26	22.74	16	84.19	27.26	64	82.54	26.87
16	82.78	23.34	32	84.68	25.17	128	82.19	26.38
20	84.91	25.61	64	86.29	27.36	256	85.41	25.08
24	86.29	27.36	128	84.87	23.26	512	86.29	27.36
32	84.13	22.89	256	83.54	26.54	1024	84.54	24.45

TABLE VIII
SCALE OF VISION-LANGUAGE MODELS (VL-MODELS) AND THEIR
IMPACT ON FINE-GRAINED WSVAD

VL-Model Variant	Model Size (Parameters)	mAP@IoU=0.3(%)	AP(%)
CLIP [10]	144M	22.38	81.19
DistilBERT + CLIP	126M	24.54	83.76
ALIGN [11]	307M	25.42	84.52
FLIP [13]	238M	25.89	84.95
DistilBERT + FLIP	165M	27.71	86.29

and performance.

5) *Analysis of Computational Complexity and Model Size:* The comparison of model size and computational complexity between the Glimpse-Emphasize Network and other methods [53], [34], [55] and [14] is presented in Table IX. The results for all methods were obtained by executing their official codes on NVIDIA RTX 4070 GPU. For an equitable comparison, open-source codes were utilized to replicate existing modules, which were then integrated into the baseline. The Glimpse-Emphasize Network demonstrates a significant reduction in model size, with only 1.14 million parameters compared to the 13.27 million of MS-BSAD [53] and 24.7 million of RTFM [34], the highest among the compared methods. Additionally, the computational complexity, measured in FLOPs, for the Glimpse-Emphasize Network stands at only 0.36 GigaFLOPs. This is much lower than the 2.63 GigaFLOPs of MS-BSAD [53] and only slightly higher than the 0.29 GigaFLOPs from HL-Net [14], indicating a balance between efficiency and complexity. Despite its smaller size and lower computational cost, the Glimpse-Emphasize Network outperforms all others in terms of effectiveness on challenging benchmarks. The findings in Table IX confirm that the Glimpse-Emphasize Network is both lightweight and efficient.

TABLE IX
COMPARISON OF MODEL SIZE(M) AND COMPUTATIONAL
COMPLEXITY(G) OF DIFFERENT METHODS

Method	Parameters (M)	FLOPs (G)	UCF-Crime AUC(%)	XD-Violence AP(%)
MS-BSAD [53]	13.27	2.63	83.15	78.92
RTFM [34]	24.7	0.79	84.30	77.81
LA-Net [55]	2.69	0.53	83.67	79.18
HL-Net [14]	5.66	0.29	83.12	79.58
Glimpse-Emphasize Network	1.14	0.36	85.76	82.44

6) *Performance comparison on different prompt templates:* The performance comparison across different prompt templates as shown in Table X highlights the effectiveness of the proposed method, particularly the 'Reparameterized Learnable Prompt (RL-Prompt) + [CLASS]' template, which achieves

TABLE X
PERFORMANCE COMPARISON OF DIFFERENT PROMPT TEMPLATE

Prompt Template	UCF-Crime AUC(%)	XD-Violence AP(%)
[CLASS]	83.95	80.69
'a video of' + [CLASS]	85.55	81.06
'a long video of' + [CLASS]	85.63	82.59
L-Prompt + [CLASS]	86.48	84.48
RL- Prompt + [CLASS]	89.14	86.29

the highest performance metrics. This template significantly outperforms other methods in the UCF-Crime dataset with an AUC of 89.14% and in the XD-Violence dataset with an AP of 86.29%. Compared to the next best performing template, the 'Learnable Prompt (L-Prompt) + [CLASS],' the proposed prompt template shows a substantial improvement of approximately 4.66% in AUC for UCF-Crime and a remarkable 1.81% increase in AP for XD-Violence. This marked enhancement underscores the method's ability to fine-tune and optimize the feature extraction process, leading to more accurate and robust anomaly detection across varying video contexts.

7) *Effectiveness of different hyper-parameter settings:* The impact of varying temperature coefficients (τ) is presented in Table XI.

TABLE XI
PERFORMANCE COMPARISON OF DIFFERENT TEMPERATURE
COEFFICIENTS τ

τ	0.01	0.03	0.05	0.07	0.09	0.1
UCF@AUC (%)	87.34	88.78	88.56	89.14	88.12	87.64
XD@AP (%)	85.33	84.45	85.13	86.29	85.20	85.36

The optimal performance is achieved with (τ) of 0.07, yielding an AUC of 89.14% on UCF-Crime and AP of 86.29% on XD-Violence. This configuration notably outperforms other tested settings, showing a remarkable increase in model efficacy. Specifically, the (τ) value of 0.07 shows an increase of up to 2.06% in AUC and 2.18% in AP over the lowest-performing configurations.

Figures 8 and 9 demonstrate the impact of the loss weight adjustment factors, λ_1 and λ_2 , by $1 \times e^{-x}$, on the performance of the proposed method across the UCF-Crime and XD-Violence datasets. The evaluation reveals that the optimal performance for λ_1 on the UCF-Crime dataset is achieved at $x = 2$, where the AUC reaches approximately 88%, and on the XD-Violence dataset at $x = 4$, achieving an AP of about 86%. Similarly, λ_2 exhibits peak performance on the

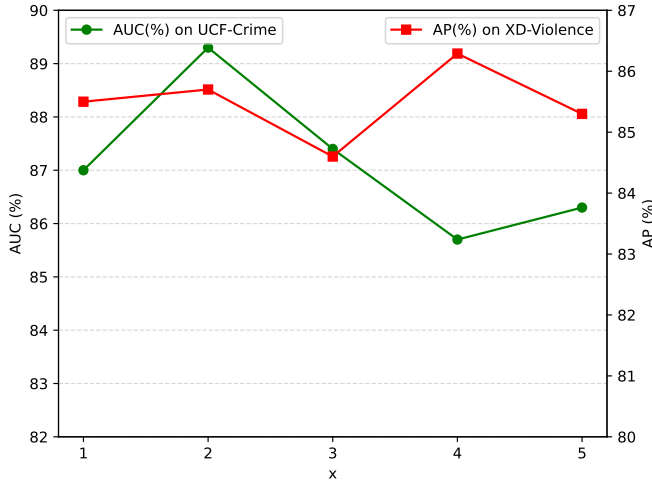


Fig. 8. Model performance under different loss weight adjustment factor λ_1 . It shows the AUC results on UCF-Crime and AP results on XD-Violence of the proposed method for different values of $\lambda_1(1 \times e^{-x})$ where, different values of x are represented as x-axis.

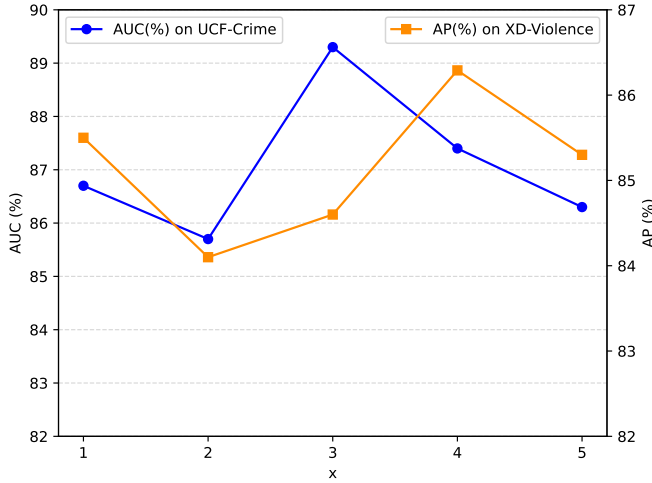


Fig. 9. Model performance under different loss weight adjustment factor λ_2 . It shows the AUC results on UCF-Crime and AP results on XD-Violence of the proposed method for different values of $\lambda_2(1 \times e^{-x})$ where, different values of x are represented as x-axis.

UCF-Crime dataset at $x = 3$ with an AUC of 89%, and the AP peaks at $x = 4$ for the XD-Violence dataset. Both figures indicate a subsequent decline in model performance as x exceeds these optimal values. This behavior highlights the sensitivity of the model to the tuning of λ values, emphasizing the necessity of precise modulation of these parameters to enhance the overall effectiveness of the anomaly detection capabilities across varied scenarios.

8) *Effectiveness of class-wise AUC and AP results* : To showcase the detailed performance on specific anomalous events, the class-wise Anomaly-Area Under Curve (Ano-AUC) results on the UCF-Crime dataset for top three performing methods: DMU, UB-MIL, and ReFLIP-VAD is illustrated in Figure 10. While ReFLIP-VAD demonstrates superior performance across most categories in the UCF-Crime dataset, there are certain failure cases where its effectiveness is re-

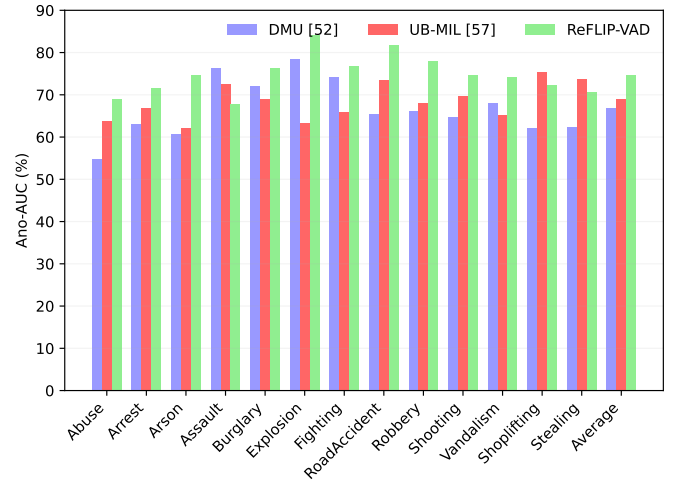


Fig. 10. Class-wise Ano-AUC results of top three methods on UCF-Crime dataset

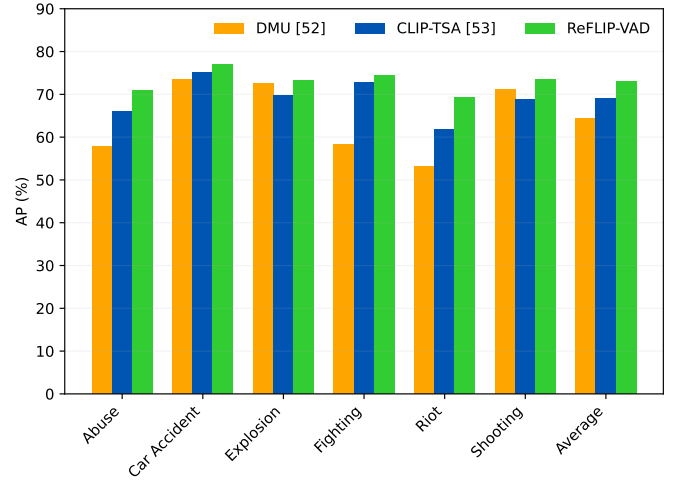


Fig. 11. Class-wise AP results of top three methods on XD-Violence dataset

duced. These shortcomings are evident in categories where the anomalies are more subtle or difficult to describe with simple textual prompts. For example, in the ‘Shoplifting’ category, ReFLIP-VAD achieves an Ano-AUC of 62.2%, which is comparable to the performance of UB-MIL at 62.3% and slightly higher than the performance of DMU at 62.2%. Similarly, in the ‘Fighting’ category, ReFLIP-VAD scores 73.5%, which, although higher than the performance of UB-MIL at 65.9% and DMU at 65.4%, is not as significantly superior as in other categories. These cases highlight the challenges ReFLIP-VAD faces in scenarios where the anomalous actions are subtle, less visually distinctive, or not easily captured by the prompt dictionary used to describe anomalies.

The class-wise Average Precision (AP) results on the XD-Violence dataset for top three performing methods: DMU, CLIP-TSA, and ReFLIP-VAD is presented in Figure 11. ReFLIP-VAD consistently demonstrates superior performance across various violence categories. For instance, in the ‘Abuse’ category, ReFLIP-VAD achieves an AP of 70.9%, outperforming CLIP-TSA and DMU, which attain 65.9% and 57.8%, respectively. Similarly, in the ‘Car Accident’ category, ReFLIP-

VAD leads with an AP of 77.1%, followed by CLIP-TSA at 75.2% and DMU at 73.4%. These results highlight the effectiveness of ReFLIP-VAD in accurately identifying and classifying different types of violent events, which is crucial for applications in surveillance and security.

V. CONCLUSION

This paper introduces ReFLIP-VAD, an innovative framework for video anomaly detection that leverages the strengths of vision-language models. By employing a reparameterized prompt encoder, ReFLIP-VAD generates contextually rich and interpretable prompt templates, enhancing the detection of specific semantics associated with anomalies. The proposed approach incorporates a dual-block architecture, consisting of a classification block and a video-text alignment block, enabling effective anomaly detection at both coarse and fine-grained levels. This approach is further strengthened by the Glimpse-Emphasize network, which captures both global and local temporal dependencies, and the MIL-Align mechanism, optimizing visual-language alignment under weak supervision. Extensive evaluations on benchmark datasets demonstrate that ReFLIP-VAD outperforms existing methods, achieving state-of-the-art performance. As a future research direction, the proposed approach can be adapted to handle situations where anomalies are not predefined, improving its effectiveness in real-world settings where unexpected events frequently occur. This will enable the model to identify and categorize new unseen types of anomalies that were not part of its initial training.

REFERENCES

- [1] A. Acsintoae, A. Florescu, M.-I. Georgescu, T. Mare, P. Sumedrea, R. T. Ionescu, F. S. Khan, and M. Shah, "Ubnorm: New benchmark for supervised open-set video anomaly detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 20 143–20 153.
- [2] Y. Liang, J. Zhang, S. Zhao, R. Wu, Y. Liu, and S. Pan, "Omni-frequency channel-selection representations for unsupervised anomaly detection," *IEEE Transactions on Image Processing*, 2023.
- [3] Z. Yang, Y. Guo, J. Wang, D. Huang, X. Bao, and Y. Wang, "Towards video anomaly detection in the real world: A binarization embedded weakly-supervised network," *IEEE Transactions on Circuits and Systems for Video Technology*, 2023.
- [4] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3d convolutional networks," in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 4489–4497.
- [5] B. Zhang and J. Xue, "Weakly-supervised anomaly detection with a sub-max strategy," *Neurocomputing*, p. 126770, 2023.
- [6] J. Carreira and A. Zisserman, "Quo vadis, action recognition? a new model and the kinetics dataset," in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4724–4733.
- [7] P. P. Dev, P. Das, and R. Hazari, "Msdeepnet: A novel multi-stream deep neural network for real-world anomaly detection in surveillance videos," in *International Conference on Deep Learning Theory and Applications*. Springer, 2023, pp. 157–172.
- [8] H. Yuan, Z. Cai, H. Zhou, Y. Wang, and X. Chen, "Transanomaly: Video anomaly detection using video vision transformer," *IEEE Access*, vol. 9, pp. 123 977–123 986, 2021.
- [9] W. Sultani, C. Chen, and M. Shah, "Real-world anomaly detection in surveillance videos," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 6479–6488.
- [10] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*. PMLR, 2021, pp. 8748–8763.
- [11] C. Jia, Y. Yang, Y. Xia, Y.-T. Chen, Z. Parekh, H. Pham, Q. Le, Y.-H. Sung, Z. Li, and T. Duerig, "Scaling up visual and vision-language representation learning with noisy text supervision," in *International conference on machine learning*. PMLR, 2021, pp. 4904–4916.
- [12] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, "Learning to prompt for vision-language models," *International Journal of Computer Vision*, vol. 130, no. 9, pp. 2337–2348, 2022.
- [13] L. Yao, R. Huang, L. Hou, G. Lu, M. Niu, H. Xu, X. Liang, Z. Li, X. Jiang, and C. Xu, "Filip: Fine-grained interactive language-image pre-training," *arXiv preprint arXiv:2111.07783*, 2021.
- [14] P. Wu, X. Liu, and J. Liu, "Weakly supervised audio-visual violence detection," *IEEE Transactions on Multimedia*, 2022.
- [15] C. Ju, T. Han, K. Zheng, Y. Zhang, and W. Xie, "Prompting visual-language models for efficient video understanding," in *European Conference on Computer Vision*. Springer, 2022, pp. 105–124.
- [16] P. Wu, X. Zhou, G. Pang, L. Zhou, Q. Yan, P. Wang, and Y. Zhang, "Vadclip: Adapting vision-language models for weakly supervised video anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 6, 2024, pp. 6074–6082.
- [17] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019.
- [18] P. Wu, J. Liu, and F. Shen, "A deep one-class neural network for anomalous event detection in complex scenes," *IEEE transactions on neural networks and learning systems*, vol. 31, no. 7, pp. 2609–2622, 2019.
- [19] Y. Liu, J. Liu, J. Lin, M. Zhao, and L. Song, "Appearance-motion united auto-encoder framework for video anomaly detection," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 69, no. 5, pp. 2498–2502, 2022.
- [20] Y. Liu, Z. Xia, M. Zhao, D. Wei, Y. Wang, S. Liu, B. Ju, G. Fang, J. Liu, and L. Song, "Learning causality-inspired representation consistency for video anomaly detection," in *Proceedings of the 31st ACM International Conference on Multimedia*, 2023, pp. 203–212.
- [21] H. Song, C. Sun, X. Wu, M. Chen, and Y. Jia, "Learning normal patterns via adversarial attention-based autoencoder for abnormal event detection in videos," *IEEE Transactions on Multimedia*, vol. 22, no. 8, pp. 2138–2148, 2019.
- [22] X. Wang, Z. Che, B. Jiang, N. Xiao, K. Yang, J. Tang, J. Ye, J. Wang, and Q. Qi, "Robust unsupervised video anomaly detection by multipath frame prediction," *IEEE transactions on neural networks and learning systems*, vol. 33, no. 6, pp. 2301–2312, 2021.
- [23] P. Jin, L. Mou, G.-S. Xia, and X. X. Zhu, "Anomaly detection in aerial videos with transformers," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–13, 2022.
- [24] Z. Yang, J. Liu, Z. Wu, P. Wu, and X. Liu, "Video event restoration based on keyframes for video anomaly detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 14 592–14 601.
- [25] Y. Chang, Z. Tu, W. Xie, B. Luo, S. Zhang, H. Sui, and J. Yuan, "Video anomaly detection with spatio-temporal dissociation," *Pattern Recognition*, vol. 122, p. 108213, 2022.
- [26] C. Guo, H. Wang, Y. Xia, and G. Feng, "Learning appearance-motion synergy via memory-guided event prediction for video anomaly detection," *IEEE Transactions on Circuits and Systems for Video Technology*, 2023.
- [27] G. Wang, Y. Wang, J. Qin, D. Zhang, X. Bao, and D. Huang, "Video anomaly detection by solving decoupled spatio-temporal jigsaw puzzles," in *European Conference on Computer Vision*. Springer, 2022, pp. 494–511.
- [28] Y. Zhong, X. Chen, J. Jiang, and F. Ren, "A cascade reconstruction model with generalization ability evaluation for anomaly detection in videos," *Pattern Recognition*, vol. 122, p. 108336, 2022.
- [29] X. Zeng, Y. Jiang, W. Ding, H. Li, Y. Hao, and Z. Qiu, "A hierarchical spatio-temporal graph convolutional neural network for anomaly detection in videos," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 33, no. 1, pp. 200–212, 2021.
- [30] Y. Zhong, X. Chen, Y. Hu, P. Tang, and F. Ren, "Bidirectional spatio-temporal feature learning with multiscale evaluation for video anomaly detection," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 32, no. 12, pp. 8285–8296, 2022.
- [31] D. Li, X. Nie, R. Gong, X. Lin, and H. Yu, "Multi-branch gan-based abnormal events detection via context learning in surveillance videos," *IEEE Transactions on Circuits and Systems for Video Technology*, 2023.
- [32] J.-X. Zhong, N. Li, W. Kong, S. Liu, T. H. Li, and G. Li, "Graph convolutional label noise cleaner: Train a plug-and-play action classifier

- for anomaly detection,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 1237–1246.
- [33] H. Lv, C. Zhou, Z. Cui, C. Xu, Y. Li, and J. Yang, “Localizing anomalies from weakly-labeled videos,” *IEEE transactions on image processing*, vol. 30, pp. 4505–4515, 2021.
- [34] Y. Tian, G. Pang, Y. Chen, R. Singh, J. W. Verjans, and G. Carneiro, “Weakly-supervised video anomaly detection with robust temporal feature magnitude learning,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 4975–4986.
- [35] C. Huang, C. Liu, J. Wen, L. Wu, Y. Xu, Q. Jiang, and Y. Wang, “Weakly supervised video anomaly detection via self-guided temporal discriminative transformer,” *IEEE Transactions on Cybernetics*, 2022.
- [36] Y. Liu, J. Liu, M. Zhao, S. Li, and L. Song, “Collaborative normality learning framework for weakly supervised video anomaly detection,” *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 69, no. 5, pp. 2508–2512, 2022.
- [37] Y. Chen, Z. Liu, B. Zhang, W. Fok, X. Qi, and Y.-C. Wu, “Mgfn: Magnitude-contrastive glance-and-focus network for weakly-supervised video anomaly detection,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 1, 2023, pp. 387–395.
- [38] Y. Fan, Y. Yu, W. Lu, and Y. Han, “Weakly-supervised video anomaly detection with snippet anomalous attention,” *IEEE Transactions on Circuits and Systems for Video Technology*, 2024.
- [39] W. Yu, Y. Liu, W. Hua, D. Jiang, B. Ren, and X. Bai, “Turning a clip model into a scene text detector,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 6978–6988.
- [40] J. Lee, J. Kim, H. Shon, B. Kim, S. H. Kim, H. Lee, and J. Kim, “Uni-clip: Unified framework for contrastive language-image pre-training,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 1008–1019, 2022.
- [41] H. Xu, G. Ghosh, P.-Y. Huang, D. Okhonko, A. Aghajanyan, F. Metze, L. Zettlemoyer, and C. Feichtenhofer, “Videoclip: Contrastive pre-training for zero-shot video-text understanding,” *arXiv preprint arXiv:2109.14084*, 2021.
- [42] L. Zanella, B. Liberatori, W. Menapace, F. Poiesi, Y. Wang, and E. Ricci, “Delving into clip latent space for video anomaly recognition,” *arXiv preprint arXiv:2310.02835*, 2023.
- [43] Z. Yang, J. Liu, and P. Wu, “Text prompt with normality guidance for weakly supervised video anomaly detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 18 899–18 908.
- [44] P. Wu, X. Zhou, G. Pang, Y. Sun, J. Liu, P. Wang, and Y. Zhang, “Open-vocabulary video anomaly detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 18 297–18 307.
- [45] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” *arXiv preprint arXiv:2010.11929*, 2020.
- [46] R. Sennrich, B. Haddow, and A. Birch, “Neural machine translation of rare words with subword units,” *arXiv preprint arXiv:1508.07909*, 2015.
- [47] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever *et al.*, “Language models are unsupervised multitask learners,” *OpenAI blog*, vol. 1, no. 8, p. 9, 2019.
- [48] P. Wu, J. Liu, Y. Shi, Y. Sun, F. Shao, Z. Wu, and Z. Yang, “Not only look, but also listen: Learning multimodal violence detection under weak supervision,” in *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXX 16*. Springer, 2020, pp. 322–339.
- [49] K.-L. Li, H.-K. Huang, S.-F. Tian, and W. Xu, “Improving one-class svm for anomaly detection,” in *Proceedings of the 2003 international conference on machine learning and cybernetics (IEEE Cat. No. 03EX693)*, vol. 5. IEEE, 2003, pp. 3077–3081.
- [50] M. Hasan, J. Choi, J. Neumann, A. K. Roy-Chowdhury, and L. S. Davis, “Learning temporal regularity in video sequences,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 733–742.
- [51] J. Zhang, L. Qing, and J. Miao, “Temporal convolutional network with complementary inner bag loss for weakly supervised anomaly detection,” in *2019 IEEE International Conference on Image Processing (ICIP)*. IEEE, 2019, pp. 4030–4034.
- [52] Y. Zhu and S. Newsam, “Motion-aware feature for improved video anomaly detection,” *arXiv preprint arXiv:1907.10211*, 2019.
- [53] Y. Zhen, Y. Guo, J. Wei, X. Bao, and D. Huang, “Multi-scale background suppression anomaly detection in surveillance videos,” in *2021 IEEE International Conference on Image Processing (ICIP)*. IEEE, 2021, pp. 1114–1118.
- [54] P. Wu and J. Liu, “Learning causal temporal relation and feature discrimination for anomaly detection,” *IEEE Transactions on Image Processing*, vol. 30, pp. 3513–3527, 2021.
- [55] Y. Pu and X. Wu, “Locality-aware attention network with discriminative dynamics learning for weakly supervised anomaly detection,” in *2022 IEEE International Conference on Multimedia and Expo (ICME)*. IEEE, 2022, pp. 1–6.
- [56] S. Park, H. Kim, M. Kim, D. Kim, and K. Sohn, “Normality guided multiple instance learning for weakly supervised video anomaly detection,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023, pp. 2665–2674.
- [57] H. Zhou, J. Yu, and W. Yang, “Dual memory units with uncertainty regulation for weakly supervised video anomaly detection,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 3, 2023, pp. 3769–3777.
- [58] H. Lv, Z. Yue, Q. Sun, B. Luo, Z. Cui, and H. Zhang, “Unbiased multiple instance learning for weakly supervised video anomaly detection,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 8022–8031.
- [59] H. K. Joo, K. Vo, K. Yamazaki, and N. Le, “Clip-tsa: Clip-assisted temporal self-attention for weakly-supervised video anomaly detection,” in *2023 IEEE International Conference on Image Processing (ICIP)*. IEEE, 2023, pp. 3230–3234.
- [60] M. Cho, M. Kim, S. Hwang, C. Park, K. Lee, and S. Lee, “Look around for anomalies: weakly-supervised anomaly detection via context-motion relational learning,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 12 137–12 146.
- [61] L. Van der Maaten and G. Hinton, “Visualizing data using t-sne,” *Journal of machine learning research*, vol. 9, no. 11, 2008.



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